ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet
A8ov5UqDdHWUAr8J81BCdUdoewNAPaGAtyH7TvQTYPoM

Date: 4/30/2025
Compounds: Amoxicillin, Caffeine

New Compound Analysis

Name: Cafmoxicillin
Formula: $C_{17}H_{22}N_5O_8S$
Weight: 455.45
Drug-likeness: 0.68

Activity:
Antibacterial with CNS stimulation properties

Target Analysis

Penicillin-binding protein 1A (PBP1A)
Binding Affinity: -9.5
Essential for bacterial cell wall biosynthesis, inhibition results in bactericidal activity.

Adenosine A1 and A2A receptors
Binding Affinity: -7.2
Antagonism leads to increased neurotransmitter release and excitation in the central nervous system.

Toxicity Profile

Risk Level: Moderate
Affected Systems: Liver, Kidneys, Central Nervous System

Mitigation Suggestions:
- Monitoring liver enzymes
- Adjusting doses to manage CNS effects
- Hydration for renal protection

Pharmacokinetics

Absorption:
Well absorbed orally, with enhanced bioavailability in presence of food

Distribution:
Wide distribution, crosses blood-brain barrier

Metabolism:
Hepatic metabolism with cytochrome P450 involvement

Excretion:
Renal excretion primarily, with minor fecal elimination

Half-life: 4-6 hours

Detailed Analysis

Cafmoxicillin is a hypothetical compound created by integrating the antibacterial properties of amoxicillin with the CNS stimulatory effects of caffeine. In terms of molecular structure, it combines key moieties from both drugs to retain bactericidal activity while conferring additional therapeutic effects targeting patients with somnolence or lethargy. The amoxicillin moiety continues to inhibit bacterial cell wall synthesis by binding to PBPs, while the caffeine component acts as an adenosine receptor antagonist facilitating neurotransmitter release and improving alertness. Despite these promising effects, Cafmoxicillin carries moderate toxicity risks, primarily affecting the liver and central nervous system, due to metabolic overload and overstimulation. Pharmacokinetic studies predict acceptable absorption parameters with renal clearance. In silico models estimate a moderate drug-likeness score suggesting further optimization might be necessary. Target analysis supports dual activity with high binding affinities for PBP1A and adenosine receptors. The initial experimental and clinical data emphasize the unprecedented nature of the compound yet guides towards critical modifications and future research geared towards increased efficacy and reduced toxicity.